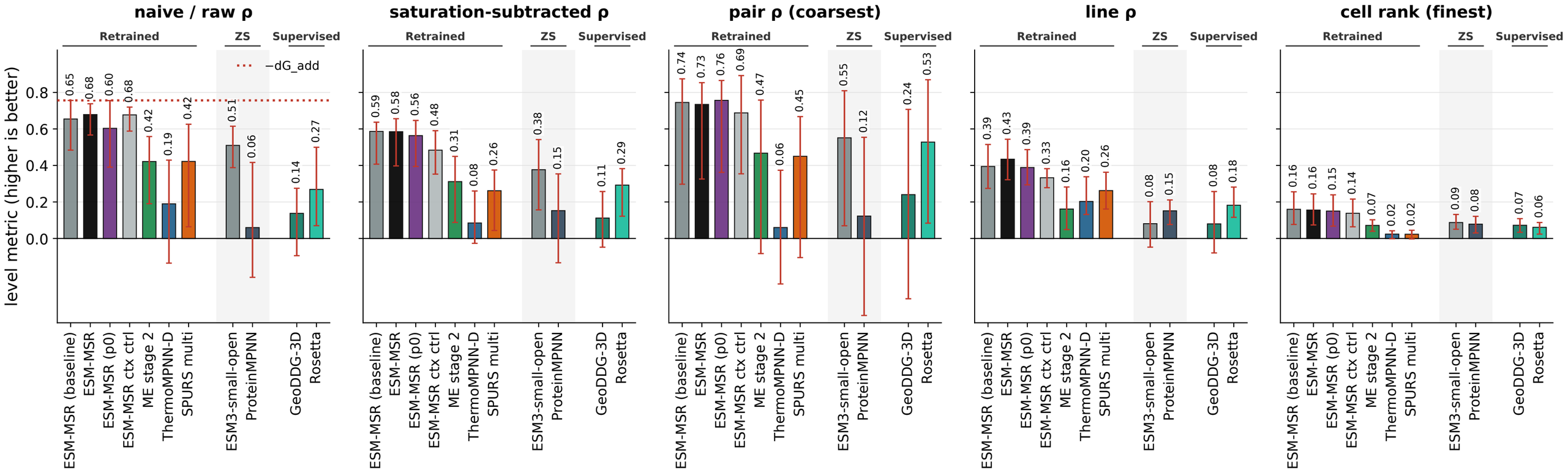
Validation: epistasis predicted level by level, 8,494 doubles with both singles measured in 9 domains



Epistasis = double-mutant prediction minus additive singles: ESM-MSR = 1/2(MT_AB+WT_AB) − WT_A − WT_B (no linking), ESM-MSR ctx ctrl = 1/2(~WT_AB+WT_AB) − WT_A − WT_B; ME stage 2, ThermoMPNN-D, SPURS multi minus the same seed's singles (3-seed means); ESM3-small-open unmasked 1/2(MT_AB+WT_AB) − WT_A − WT_B; ProteinMPNN, GeoDDG-3D, Rosetta minus sum of singles. Intervals: 95% bootstrap over domains.
Levels (residuals r = dddG − E[dddG | x] and r' = dddG' − E[dddG' | x'] via 25-quantile-bin smoothers on additive dG expectations x, x'): (1) naive / raw ρ: single Spearman ρ pooled over all doubles on raw dddG;
(2) saturation-subtracted ρ: single Spearman ρ pooled over all doubles on residuals r', r; (3) pair ρ: single Spearman ρ across position-pair matrices (≥20 doubles) of each pair's mean residual;
(4) line ρ: single Spearman ρ pooled over all row/column lines (≥3 cells) of line-mean residuals minus their pair mean; (5) cell rank: computed per position-pair complete block (≥3×3) and averaged across blocks — within each block, column ranks of ddG_AB are double-centred (row, column and block means removed) and correlated by Spearman ρ (averaged over both row/column orientations).